

- **The target task: SARCASM DETECTION**
(to predict if a text is sarcastic or not solely from the text)
- **The method: intermediate-task transfer learning**

Intermediate-Task Transfer Learning with BERT for Sarcasm Detection

(Edoardo Savini and Cornelia Caragea, 2022)

- **The idea: The BERT models are improved by fine-tuning them on related intermediate tasks (sentiment and emotion) before fine-tuning them on the target task (sarcasm detection).**

Note that:

- **Sarcasm detection has become one of the most challenging tasks in NLP because of the highly figurative nature of sarcasm.**
- **Savini and Caragea (2022) analyze the effectiveness of BERT and intermediate-task transfer learning with BERT on the sarcasm detection task and find a neural framework able to accurately predict sarcasm in many types of social platforms.**
- **Diverse intermediate tasks are used (see the paper).**
- **The performance of the BERT models is studied on three datasets (of different sizes and characteristics), collected from:**
 - ✓ **The Internet Argument Corpus (IAC)**
 - ✓ **Reddit**
 - ✓ **Twitter**

The conclusions:

- **For sarcasm detection, the improvement in performance of BERT-based models (with and without intermediate task transfer learning) is as high as 11.53%.**
- **The performance of BERT can be improved by fine-tuning on data-rich intermediate tasks, before fine-tuning the BERT models on the sarcasm detection target task.**
- **Depending on the characteristics of the target task data, different intermediate tasks are more useful than others.**
- **When using transfer learning in the case of sarcasm, sentiment works best as intermediate task.**

BASELINE MODELING

1. BERT Pre-trained Language Model

Savini and Caragea (2022) fine-tune BERT `bert-base-uncased` from the HuggingFace Transformers library on the target task (sarcasm detection) with an added single linear layer on top as a sentence classifier that uses the final hidden state corresponding to the [CLS] token.

2. Intermediate-Task Transfer Learning

Due to the strong correlation between sarcasm and (implied negative) sentiment and emotions, Savini and Caragea (2022) propose to explore transfer learning from the related intermediate tasks of *sentiment classification* and *emotion detection*.

The purpose of intermediate task transfer learning:

The main purpose of intermediate task transfer learning is to use data-rich sources of relevant information (e.g. sentiment) when the dataset for the target task (sarcasm in this case) is small in size.

The transfer learning framework



- **The target task**: sarcasm detection from textual inputs
(Given a piece of text, predict if the text is sarcastic or not, solely from the text).

- **The intermediate tasks**:

1. **Fine-Grained EmoNET**

- ❖ **EmoNet** is a Twitter dataset composed of tweets automatically annotated with the Plutchik emotion set. Savini and Caragea (2022) used a smaller version that contains 50000 tweets annotated with the Plutchik-8 emotion set (joy, surprise, trust, anticipation, sadness, fear, anger, disgust).
- ❖ **BERT** was fine-tuned on the EmoNet tweets in a supervised fashion before fine-tuning it on the sarcasm detection target task.

2. Course-Grained EmoNet

- ❖ **Same dataset as before except the labels for each tweet is made more abstract, according to the polarized emotions (positive and negative). We call this dataset EmoNetSent.**
- ❖ **All the emotion labels with negative insight (sadness, fear, anger, disgust) are grouped into a negative sentiment label (0); the remaining emotions (joy, surprise, trust, anticipation) are grouped into a positive class.**
- ❖ **BERT is fine-tuned on EmoNetSent in a supervised fashion before fine-tuning it on the sarcasm detection target task.**

3. IMDB Movie Review

- ❖ A sentiment dataset created for learning word vector representations to capture semantic information from text.**
- ❖ Contains 50,000 polarized movie reviews labeled with binary sentiment classes (positive, negative).**

COMPARISON

- **IMDB and EmoNet differ in terms of text length and content:**
 - **EmoNet tweets contain short-length sentences explicitly dense of emotional charge;**
 - **IMDB consists of very long phrases and sentences in which the sentiment (lexical) content appears more implicit and sparse along the sentences.**

3. Standard Neural Model

➤ The baselines for evaluation:

- **BiLSTM (Bidirectional Long Short Term Memory) models – perform well on text classification.**
- **CNN (Convolutional Neural Networks) – perform well on sentence classification tasks.**

➤ Note that:

Hyperparameter settings are taken from Kim, Y., Convolutional Neural Networks for Sentence Classification. In: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP), Doha, Qatar, 25-29 October 2014; pp. 1746-1751.

4. Data

- **The Sarcasm V2 Corpus**
- **SARC (the Self-Annotated Reddit Corpus)**
- **A dataset from Twitter (SARCTwitter)**

➤ **The Sarcasm V2 Corpus (Oraby et al., 2016):**

- ✓ **small size dataset;**
- ✓ **contains 4692 lines with sentences from dialogue examples on political debates from the Internet Argument Corpus (IAC 2.0).**

- **SARC - the Self-Annotated Reddit Corpus (Khodak et al., 2018):**
 - ✓ **large-scale self-annotated corpus for sarcasm;**
 - ✓ **SARC contains more than a million sarcastic and non-sarcastic statements retrieved from Reddit (a social media site) with some contextual information;**
 - ✓ **almost every comment is made of one sentence;**
 - ✓ **annotation is performed by the authors of the comment themselves;**
 - ✓ **SARC has many variants; here the experiments use only the first version of the “Main Balanced dataset”, equally composed of 505,413 sarcastic and 505,413 non-sarcastic statements;**
 - ✓ **total train size: 1,010,826 statements.**
 - ✓ **The authors also provide a balanced test set of 251,608 comments – which are used for model evaluation.**

➤ **SARCTwitter** (Mishra et al., 2016):

- ✓ dataset having a similar structure to the one used by the intermediate task; these comments have a structure similar to the EmoNet ones;
- ✓ dataset consisting of 994 tweets from Twitter, manually annotated with both sarcastic and sentiment information;
- ✓ each tweet has two labels, one for sentiment and one for sarcasm;
- ✓ out of 994 tweets, 383 are labeled as positive (sentiment) and 611 are labeled as negative (sentiment);
- ✓ out of the 994 tweets, 350 are labeled as sarcastic and 644 are labeled as non-sarcastic.

➤ **Note that:**

- **The SARCTwitter dataset contains also eye-movement of the readers, that was ignored during the experiments, since the task is to detect sarcasm *solely* from the text content.**

5. Experiments

- For every epoch, Savini and Caragea (2022) compute F1-score and Accuracy.
- The training is stopped (for both target task and intermediate tasks) once the average F1 on the validation set ceases to grow after some consecutive epochs (the patience is set to 5).
- For all the performed experiments and the implementation details, see the paper Savini, E., Caragea, C. Intermediate-Task Transfer Learning with BERT for Sarcasm Detection. *Mathematics* 2022, 10(5), 844;
<https://doi.org/10.3390/math10050844>

6. RESULTS

6.1. Results on Sarcasm V2 Corpus

- **All the experiments of Savini and Caragea (2022) outperform the existing state-of-the-art for this corpus, establishing new baselines (see the tables in the paper).**
- **The BiLSTM and CNN baselines, proposed by Savini and Caragea (2022) obtain similar performance with each other and exceed the previous state-of-the-art by 2%.**
- **The same BiLSTM and CNN baselines are outperformed by the simple BERT model by 4%.**
- **These results prove the efficacy of neural models with word embeddings over feature engineering methods with SVMs.**
- **The transfer models, except for TransferEmoNet reach similar results (see Table 10 in the paper).**

❖ Other Comments:

- The lower performance of TransferEmoNet can be explained by the scarce emotion distribution in IAC (Internet Argument Corpus). Also: the Sarcasm V2 comments are mainly responses to debates, in which emotions such as “fear” and “anticipation” are very rare.
- The model pre-trained on the IMDB dataset achieves state-of-the-art performance. It outperforms the vanilla BERT model by 0.3%. (This increase may be explained by the fact that the features of the Sarcasm V2 comments are more similar to the ones of movie reviews rather than tweets, which are shorter).
- The expressions of sentiment/emotions are more obvious in EmoNet compared with IMDB. Thus the examples in the IMDB dataset are more challenging to the model and therefore more beneficial for learning.

❖ **Final conclusions corresponding to the Sarcasm V2 Corpus:**

- **BERT outperforms previous approaches on this dataset and acts as a strong baseline.**
- **There is correlation between sarcasm and sentiment.**

6.2. Results on SARC

- **All models proposed by Savini and Caragea (2022) outperform all other prior works by at least 10% (see Table 11 in the paper).**
- **The efficacy of capturing semantics through the pre-trained language models for the sarcasm prediction task is confirmed.**
- **Even the simplest model trained with BERT Base outperforms all the previous works. The transfer-learning improves the performance of BERT base model only slightly.**
- **The sentiment information from EmoNet slightly improved the efficacy of the simple BERT classification, making the TransferEmoNetSent model the new state-of-the-art performance on the SARC dataset.**

❖ **Final conclusions regarding the SARC dataset:**

- **BERT appears as the best-performing model.**
- **The importance of sentiment in sarcasm classification is underlined.**

6.3. Results on SARCTwitter

- **All models proposed by Savini and Caragea (2022) outperform the previous state-of-the-art by at least 5% (see Table 12 in the paper).**
- **These results prove the strength of large pre-trained language models for this task.**
- **In particular, BERT is again shown to be the most suitable model for sarcasm detection.**
- **Unlike the previous experiments, here the transfer learning models, especially on EmoNetSent, show improvement over vanilla BERT (~1% for EmoNetSent). A potential reason for this is the size of the datasets, with SARCTwitter being smaller. (SARC has 1M examples in the training set and the models are already well trained on this dataset).**

❖ Other comments:

- This proves the main purpose of intermediate task transfer learning is to use data-rich sources of relevant information (sentiment, in this case) when the dataset for the target task (sarcasm, in this case) is small in size.
- The results validate the fact that when the dataset size is small (the SARCTwitter dataset, in this case) using BERT with intermediate-task transfer learning achieves a substantially better performance compared with vanilla BERT.
- The results again prove the correlation between sarcasm and sentiment.

❖ Final conclusions

- An intermediate task training on a correlated task can improve the effectiveness of the base BERT models.
- Sentiment as intermediate task has a higher impact than emotions on the performance, especially on sarcasm detection datasets that are small in size.

❖ Future study

- Study the impact of integrating multiple intermediate tasks at the same time!